Scientific Visualization

Assignment 5

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Solutions:

Drawing arrow/glyphs

1- If there is no data loaded exit the function
2- Define the constants in NDC units
3- Compute the arrow heads rotation angles
4- iterate over grid with sampling_rate stride
    a. indexing for the grid idx = (loaded_timestep * data_size + j * W + i) * 3
    b. calculate the magnitude =sqrt[(vx*vx)+ (vy*vy)]
    c. skip near zero vector if magnitude less than 1e^-6
    d. calculate unit direction dx= vx /magnitude , dy = vy / magnitude
    e. choose arrow length depends on the user choice
    f. convert the grid (i,j) to NDC position
    g. compute tip: px = px+L *dx
    h. compute tip: py = py+L* dy
    i. compute arrow head wings
    j. push all line segments to VBO
5- draw line segments

streamLines Algorithm

Inputs:

Initial position (x, y)

integration step size dt

integration method (Euler or RK2)

maximum streamline length

maximum step count

1) before start running the algorithm, check if the data is loaded , if not return
2) do backward integration with the point the user click on it with d = -1.....
    a. start the path with pushing the initial point (seed) to the list
    b. loop through predefined maximum step:
        i. sample bilinear for the point p
        ii. if the point is less than Epsilon(1e^-5)
            1. stop here
        iii. sample bilinear for p.x, p.y and save it to v1
        iv. if selected to use Euler
            1. the new position = p + direction * dt *v1
        v. else RK2
            1. calculate the mid point = p + direction * dt * 0.5 * v1
            2. sample bilinear for mid.x, mid.y and save it v2
            3. the new position = p + direction * dt * v2
        vi. check the bound for the new position, if the new position from euler or RK2 length, if it is out of bound
            1. stop here
        vii. calculate the new total of the stramline length

viii. if the new total of the length is larger than a predefined maximum_length
    1. stop here
ix. continue to the next point

3) do forward integration with the point the user click on it p with d= +1,.....
    a. same steps for forward but with different directions
4) concatenate the reversed backward path with the forward path and insert it in the list after converting it to Normalized Device Coordinates NDC
5) check if the size of list more than or equal to 4(each vertex is two floats, more than 2 vertex) if yes add it to streamlines list for drawing

pathLines Algorithm

input:

position (x,y)

starting timestep t

integration step size dt

integration method (Euler or K2)

Maximum length

Maximum step count

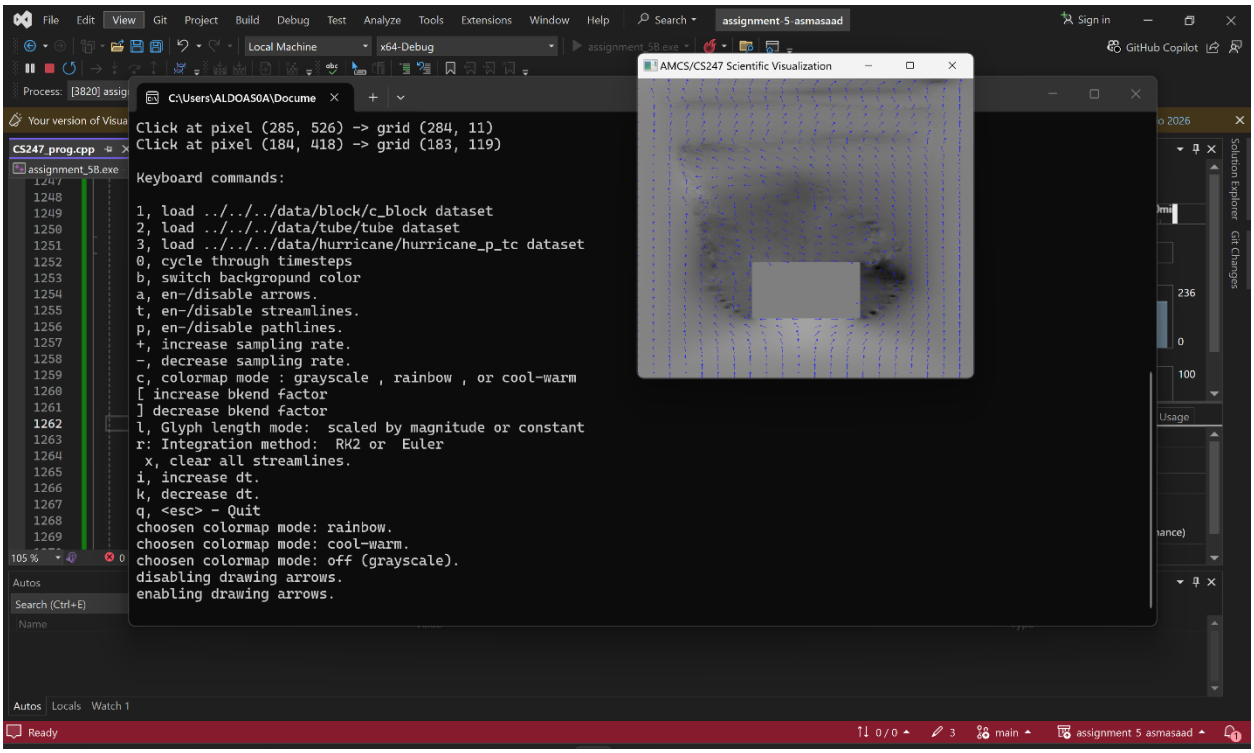
1) before start running the algorithm, check if the data is loaded, if not return
2) check if time steps less than or equal to one , if yes return
3) do backward integration with the point the user click on (x,y), and the starting time t , with direction parameter d =-1
    a. start the path with pushing the initial point (seed) to the list
    b. loop through predefined maximum step:
        i. sample Trilinear for the point p and the time step t
        ii. if the point is less than Epsilon(1e^-5)
            1. stop here

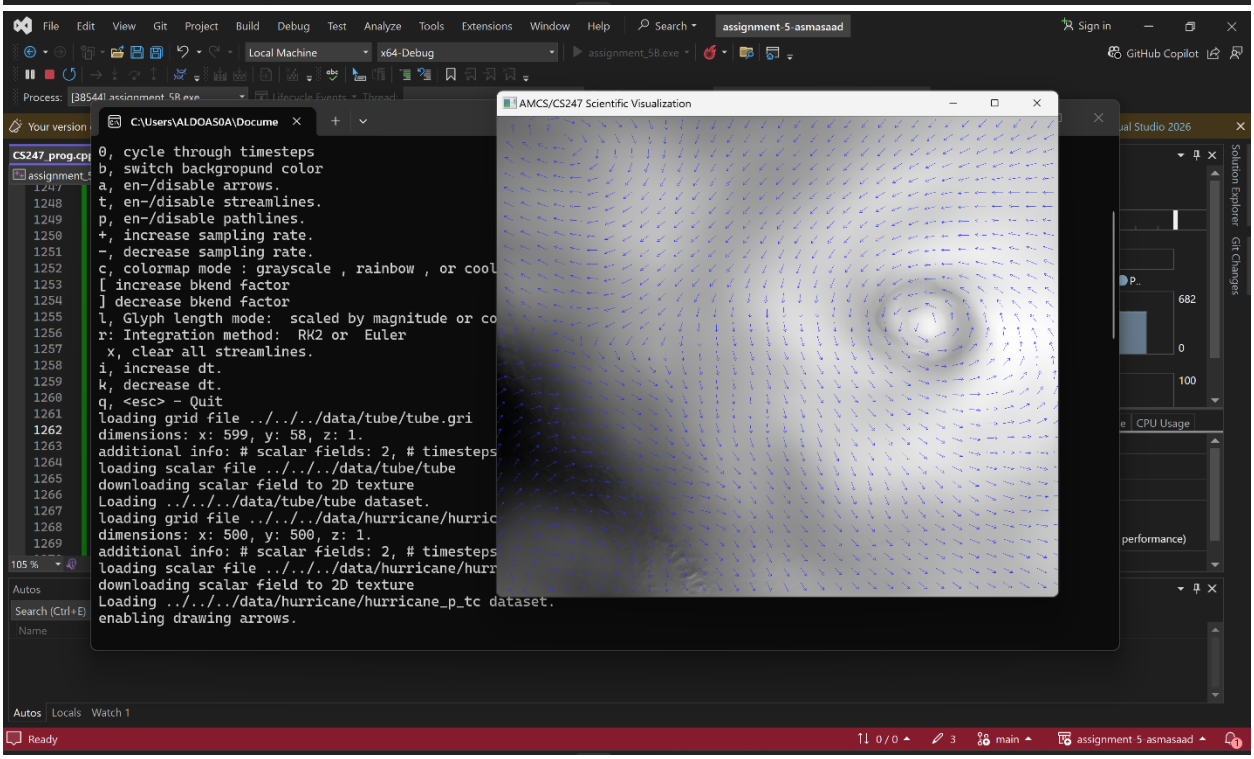
iii. sample trilinear for p.x, p.y, and time step t and save it to v1
iv. if selected to use Euler
    1. the new position = p + direction * dt *v1
v. else RK2
    1. calculate the mid point = p + direction * dt * 0.5 * v1
    2. calculate the mid point in time = t + direction * dt * 0.5f
    3. sample trilinear for mid.x, mid.y , mid_t and save it v2
    4. the new position = p + direction * dt * v2
vi. calculate the new time = t + direction * dt
vii. check the bound for the new position, if the new position from euler or RK2 length, if it is out of bound
    1. stop here
viii. check if the new time is less than zero or larger or equal num_timesteps
    1. stop here
ix. calculate the new total of the streamline length
x. if the new total of the length is larger than a predefined maximum_length
    1. stop here
xi. continue to the next point & the next time

4) do forward integration with the point the user click on (x,y), and the starting time t , with direction parameter d =+1
    a. same steps for forward but with different directions
5) concatenate all the point coming after forward integration and backward integration in the list after converting it to Normalized Device Coordinates NDC
6) check if the size of list is more than or equal to 4(each vertex is two floats, more than 2 vertex) if yes add it to streamlines list for drawing

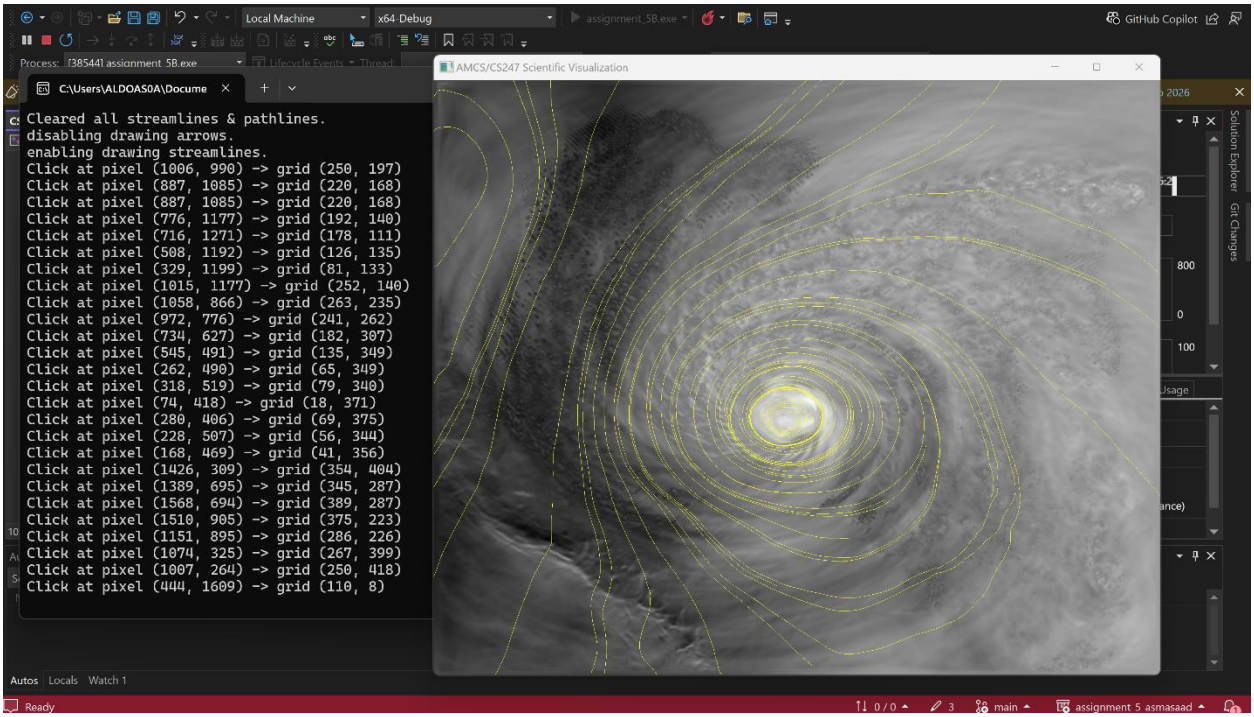
Screenshots:

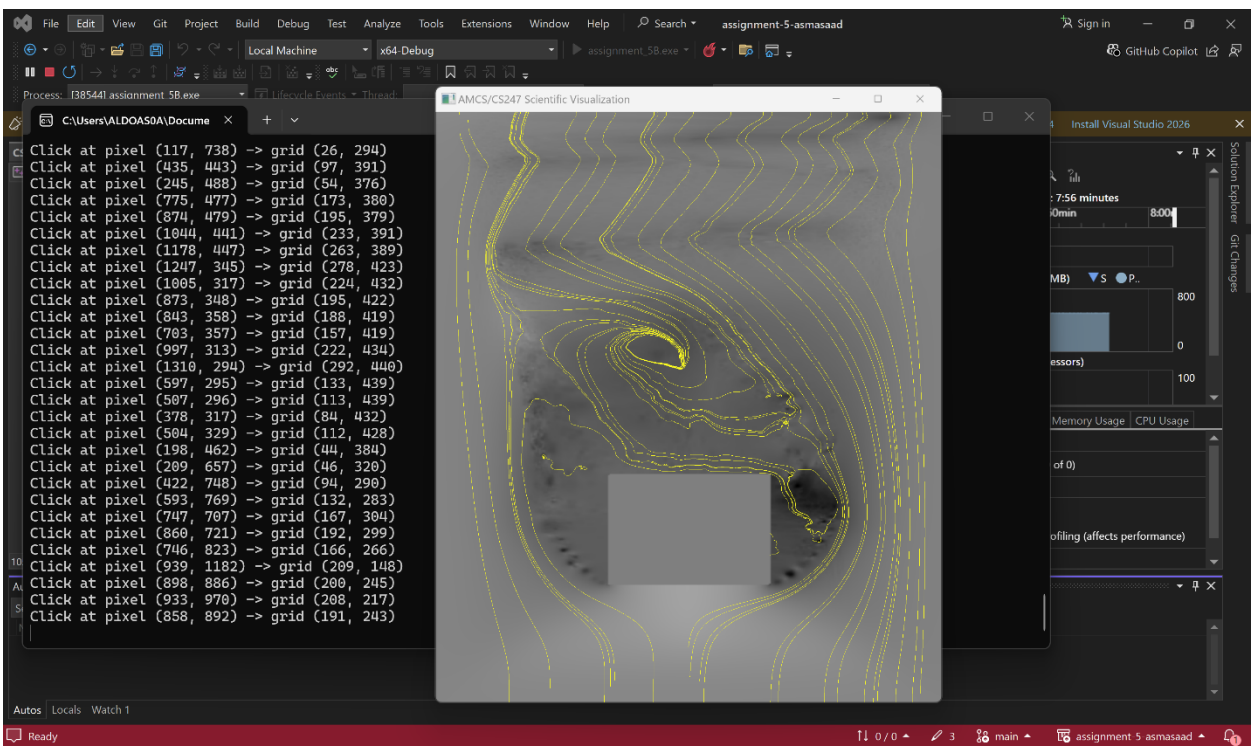
Glyphs:


AMCS/CS247 Scientific Visualization
Click at pixel (285, 526) -> grid (284, 11)
Click at pixel (184, 418) -> grid (183, 119)
Keyboard commands:
1, load ../../../data/block/c_block dataset
2, load ../../../data/tube/tube dataset
3, load ../../../data/hurricane/hurricane_p_tc dataset
0, cycle through timesteps
b, switch backgropund color
a, en-/disable arrows.
t, en-/disable streamlines.
p, en-/disable pathlines.
+, increase sampling rate.
-, decrease sampling rate.
c, colormap mode : grayscale , rainbow , or cool-warm
[increase bkend factor
] decrease bkend factor
l, Glyph length mode: scaled by magnitude or constant
r: Integration method: RK2 or Euler
x, clear all streamlines.
i, increase dt.
k, decrease dt.
q, <esc> - Quit
choosen colormap mode: rainbow.
choosen colormap mode: cool-warm.
choosen colormap mode: off (grayscale).
disabling drawing arrows.
enabling drawing arrows.

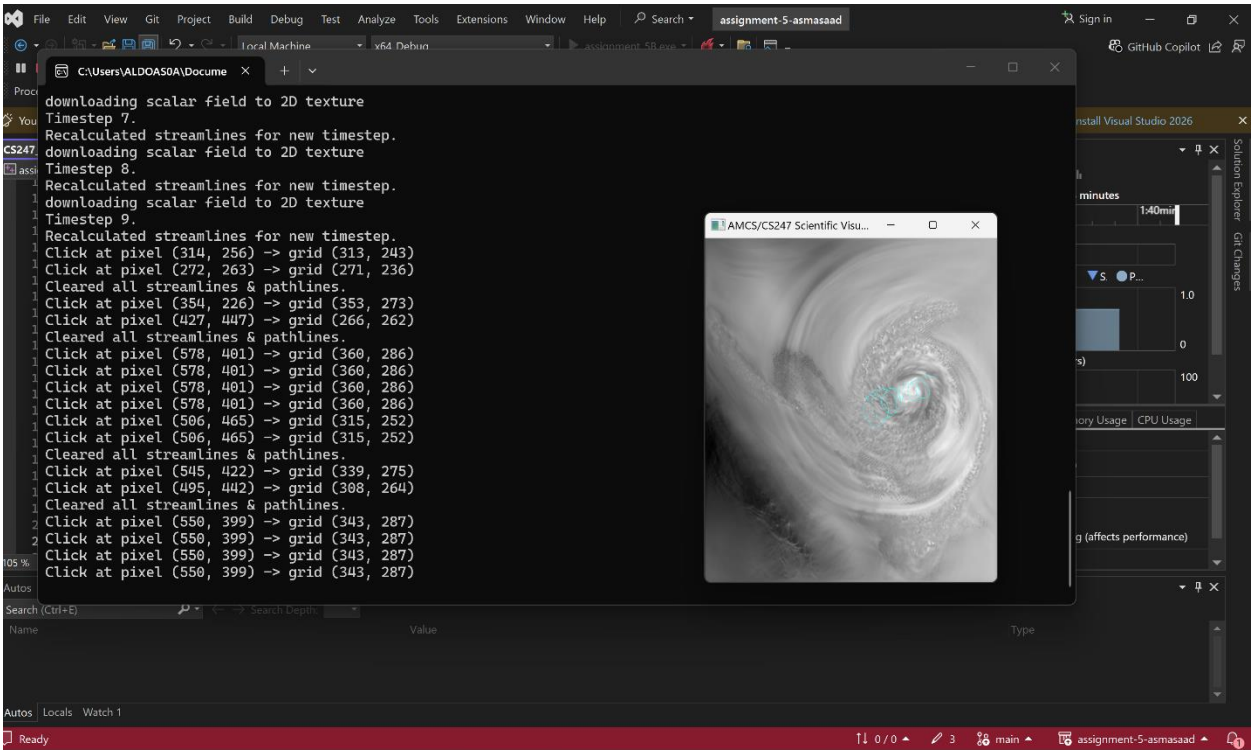

AMCS/CS247 Scientific Visualization
0, cycle through timesteps
b, switch backgropund color
a, en-/disable arrows.
t, en-/disable streamlines.
p, en-/disable pathlines.
+, increase sampling rate.
-, decrease sampling rate.
c, colormap mode : grayscale , rainbow , or cool
[increase bkend factor
] decrease bkend factor
l, Glyph length mode: scaled by magnitude or co
r: Integration method: RK2 or Euler
x, clear all streamlines.
i, increase dt.
k, decrease dt.
q, <esc> - Quit
loading grid file ../../../data/tube/tube.gri
dimensions: x: 599, y: 58, z: 1.
additional info: # scalar fields: 2, # timesteps
loading scalar file ../../../data/tube/tube
downloading scalar field to 2D texture
Loading ../../../data/tube/tube dataset.
loading grid file ../../../data/hurricane/hurric
dimensions: x: 500, y: 500, z: 1.
additional info: # scalar fields: 2, # timesteps
loading scalar file ../../../data/hurricane/hurr
downloading scalar field to 2D texture
Loading ../../../data/hurricane/hurricane_p_tc dataset.
enabling drawing arrows.

StreamLine:

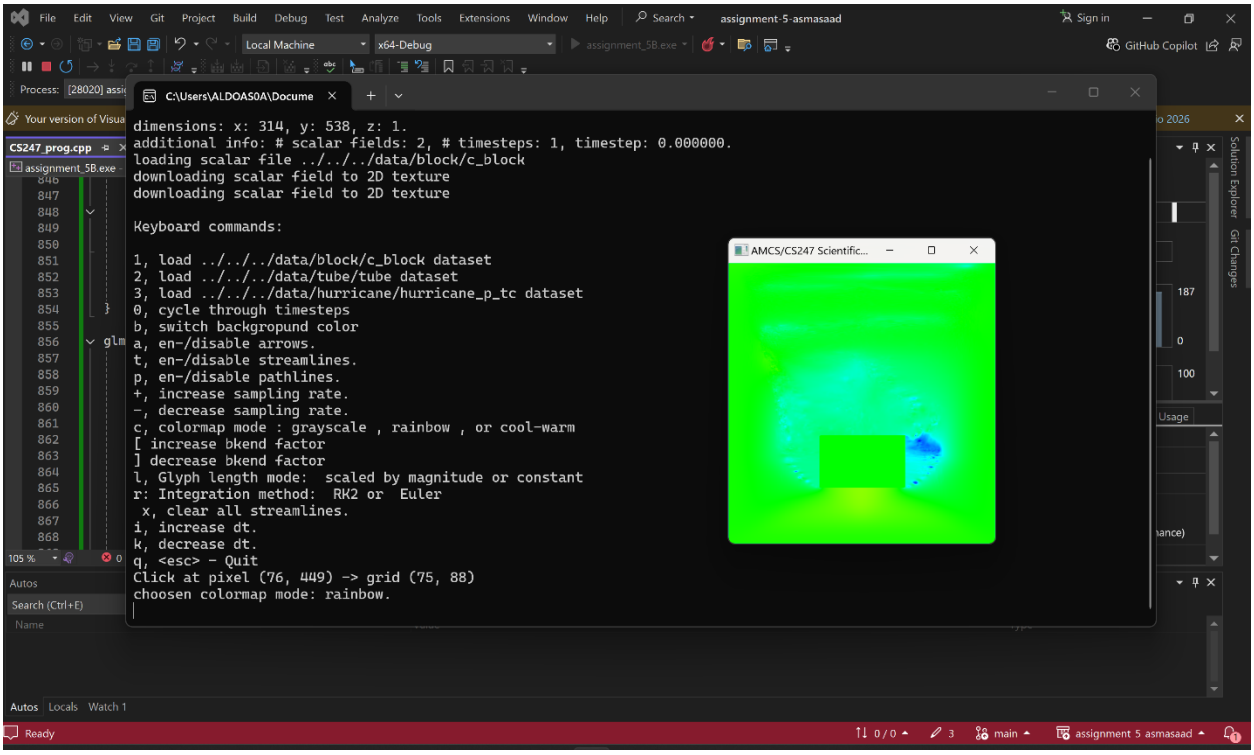

Local Machine
x64-Debug
assignment_5B.exe
GitHub Copilot
Process: [385441] assignment_5B.exe
C:\Users\ALDOAS0A\Docume
AMCS/CS247 Scientific Visualization
Cleared all streamlines & pathlines.
disabling drawing arrows.
enabling drawing streamlines.
Click at pixel (1006, 990) -> grid (250, 197)
Click at pixel (887, 1085) -> grid (220, 168)
Click at pixel (887, 1085) -> grid (220, 168)
Click at pixel (776, 1177) -> grid (192, 140)
Click at pixel (716, 1271) -> grid (178, 111)
Click at pixel (508, 1192) -> grid (126, 135)
Click at pixel (329, 1199) -> grid (81, 133)
Click at pixel (1015, 1177) -> grid (252, 140)
Click at pixel (1058, 866) -> grid (263, 235)
Click at pixel (972, 776) -> grid (241, 262)
Click at pixel (734, 627) -> grid (182, 307)
Click at pixel (545, 491) -> grid (135, 349)
Click at pixel (262, 490) -> grid (65, 349)
Click at pixel (318, 519) -> grid (79, 340)
Click at pixel (74, 418) -> grid (18, 371)
Click at pixel (280, 406) -> grid (69, 375)
Click at pixel (228, 507) -> grid (56, 344)
Click at pixel (168, 469) -> grid (41, 356)
Click at pixel (1426, 309) -> grid (354, 404)
Click at pixel (1389, 695) -> grid (345, 287)
Click at pixel (1568, 694) -> grid (389, 287)
Click at pixel (1510, 905) -> grid (375, 223)
Click at pixel (1151, 895) -> grid (286, 226)
Click at pixel (1074, 325) -> grid (267, 399)
Click at pixel (1007, 264) -> grid (250, 418)
Click at pixel (444, 1609) -> grid (110, 8)
Autos
Locals
Watch 1
Ready
main
assignment 5 asmasaad


File
Edit
View
Git
Project
Build
Debug
Test
Analyze
Tools
Extensions
Window
Help
Search
assignment-5-asmasaad
Sign in
Local Machine
x64-Debug
assignment_5B.exe
GitHub Copilot
Process: [385441] assignment_5B.exe
C:\Users\ALDOAS0A\Docume
AMCS/CS247 Scientific Visualization
Install Visual Studio 2026
Click at pixel (117, 738) -> grid (26, 294)
Click at pixel (435, 443) -> grid (97, 391)
Click at pixel (245, 488) -> grid (54, 376)
Click at pixel (775, 477) -> grid (173, 380)
Click at pixel (874, 479) -> grid (195, 379)
Click at pixel (1044, 441) -> grid (233, 391)
Click at pixel (1178, 447) -> grid (263, 389)
Click at pixel (1247, 345) -> grid (278, 423)
Click at pixel (1005, 317) -> grid (224, 432)
Click at pixel (873, 348) -> grid (195, 422)
Click at pixel (843, 358) -> grid (188, 419)
Click at pixel (703, 357) -> grid (157, 419)
Click at pixel (997, 313) -> grid (222, 434)
Click at pixel (1310, 294) -> grid (292, 440)
Click at pixel (597, 295) -> grid (133, 439)
Click at pixel (507, 296) -> grid (113, 439)
Click at pixel (378, 317) -> grid (84, 432)
Click at pixel (504, 329) -> grid (112, 428)
Click at pixel (198, 462) -> grid (44, 384)
Click at pixel (209, 657) -> grid (46, 320)
Click at pixel (422, 748) -> grid (94, 290)
Click at pixel (593, 769) -> grid (132, 283)
Click at pixel (747, 707) -> grid (167, 304)
Click at pixel (860, 721) -> grid (192, 299)
Click at pixel (746, 823) -> grid (166, 266)
Click at pixel (939, 1182) -> grid (209, 148)
Click at pixel (898, 886) -> grid (200, 245)
Click at pixel (933, 970) -> grid (208, 217)
Click at pixel (858, 892) -> grid (191, 243)
Autos
Locals
Watch 1
Ready
main
assignment 5 asmasaad

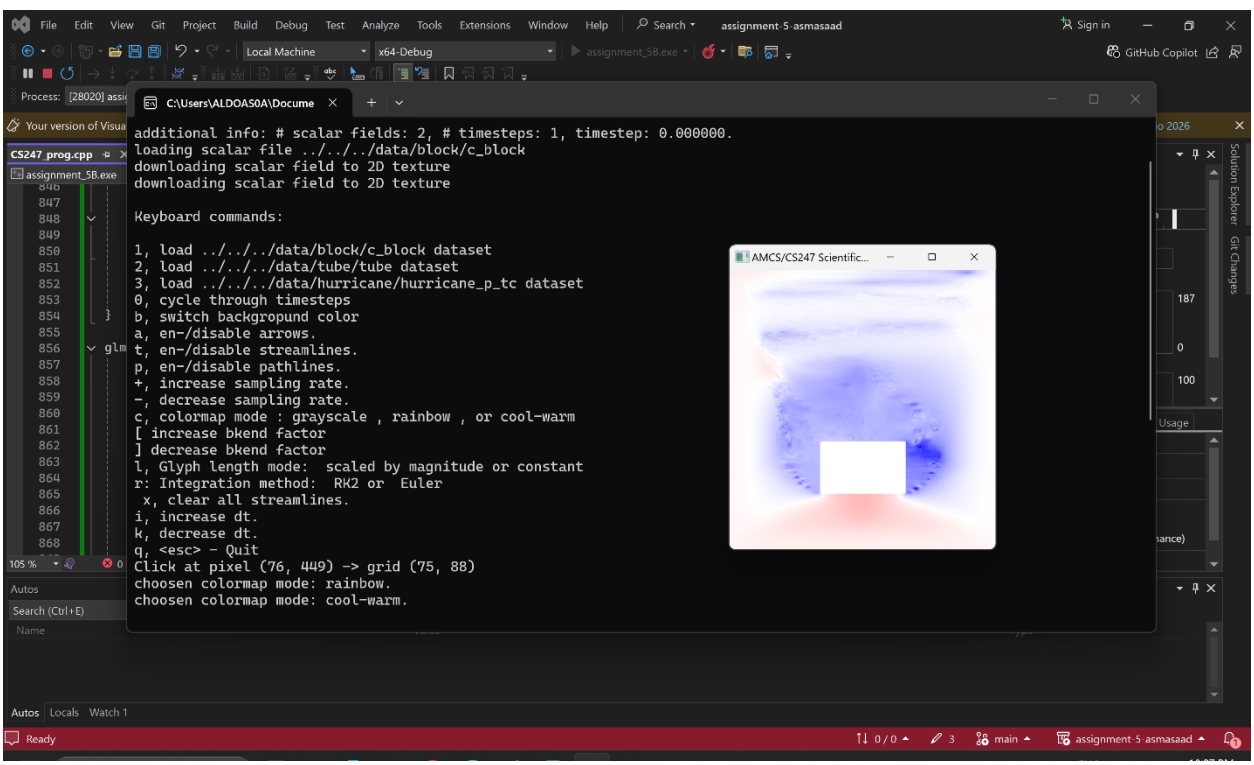
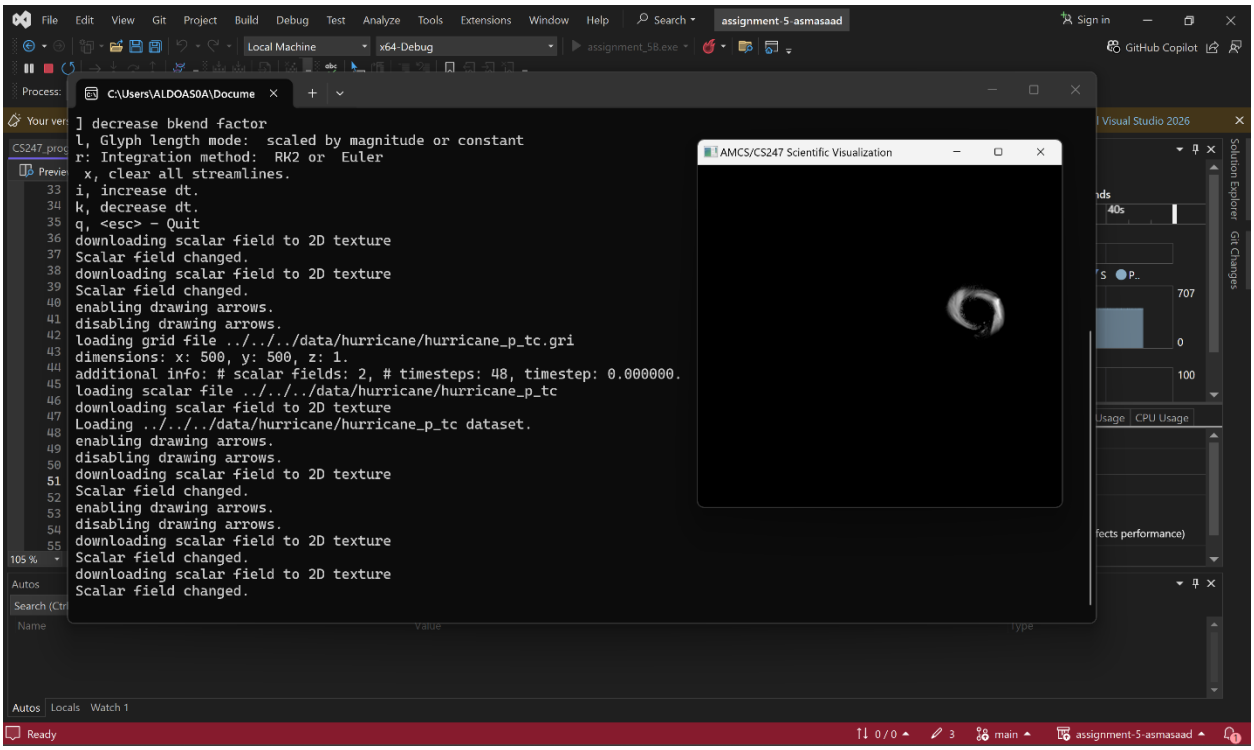
PathLine:



Rainbow color-mode:



Color-warm colormode



Scalar Field